

ENSO-Conditioned Seasonal Rainfall Extremes and Signal Preservation after Cross-Fitted Quantile Delta Mapping over Uttaradit, Thailand

SUPPLEMENTARY INFORMATION

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Scope

These tables provide the model, station, event-classification, secondary-index, inference, bias-adjustment, quality-control, and provenance details supporting the main article. Values are generated from the frozen 1981-2014 analysis outputs. Secondary-index responses are descriptive and were not included in the pre-specified primary multiple-testing family. Full row-level QDM and quality-control diagnostics remain in the machine-readable reproducibility package.

Supplementary Table S1. CMIP6 precipitation and exact-member Niño-3.4 source configuration.

Model	Member	pr grid	tos grid	tos version	Niño-3.4 cells	Unique pr grids
ACCESS-ESM1-5	r1i1p1f1	gn	gn	20191115	1500	2
CanESM5	r1i1p1f1	gn	gn	20190429	1450	1
CESM2	r1i1p1f1	gn	gn	20190514	1710	3
EC-Earth3	r1i1p1f1	gr	gn	20200918	1450	5
FGOALS-g3	r1i1p1f1	gn	gn	20191107	969	2
MIROC6	r1i1p1f1	gn	gn	20181212	1000	2
MRI-ESM2-0	r1i1p1f1	gn	gn	20190904	1683	3

Note. All model Niño-3.4 indices use area-weighted exact-member tos, a 1981-2010 monthly climatology, centred linear detrending, and a centred three-month mean. EC-Earth3 intentionally uses pr on gr and tos on gn.

Supplementary Table S2. Uttaradit station metadata, probable-missing-month corrections, and complete management seasons.

Station	Latitude	Longitude	Elevation (m)	Probable-missing months	Days removed	Complete rainy	Complete hot/dry
351001	17.23	100.10	54.6	2	61	34	32
351002	17.42	100.13	62.0	15	459	33	22
351003	17.67	100.68	371.8	6	180	34	27
351004	17.65	100.38	115.8	4	121	34	29
351005	17.60	100.05	81.5	9	273	34	25
351006	17.92	100.88	348.8	2	62	33	32
351007	18.02	101.07	427.9	0	0	34	33
351008	17.47	100.37	82.4	1	28	34	32
351009	17.75	100.28	95.3	0	0	34	33
351010	17.90	100.82	214.7	0	0	34	33
351011	17.56	100.11	65.4	0	0	34	33
351012	17.75	100.92	415.1	1	31	34	32
351201	17.62	100.10	67.0	3	93	34	30

Note. Observed zero screening was performed on the native Gregorian calendar. In total, 43 station-months (1,308 daily values) were classified as probable missing; confirmed dry and uncertain zeros were retained.

Supplementary Table S3. Observed management-season ENSO classification from frozen NOAA CPC ERSSTv6 ONI.

Climate year	Rainy phase	Rainy mean ONI (°C)	Rainy windows	Hot/dry phase	Hot/dry mean ONI (°C)	Hot/dry windows
1981	Neutral	-0.24	8	Transition/unclassified	0.07	8
1982	El Niño	1.10	8	El Niño	1.70	8
1983	Transition/unclassified	0.10	8	La Niña	-0.65	8
1984	Transition/unclassified	-0.49	8	La Niña	-0.86	8
1985	Transition/unclassified	-0.47	8	Neutral	-0.29	8
1986	Transition/unclassified	0.38	8	El Niño	1.05	8
1987	El Niño	1.24	8	Transition/unclassified	0.40	8
1988	La Niña	-1.21	8	La Niña	-1.32	8
1989	Transition/unclassified	-0.40	8	Neutral	0.12	8
1990	Neutral	0.30	8	Transition/unclassified	0.32	8
1991	El Niño	0.70	8	El Niño	1.27	8
1992	Transition/unclassified	0.39	8	Neutral	0.25	8
1993	Neutral	0.40	8	Neutral	0.15	8
1994	Transition/unclassified	0.42	8	El Niño	0.61	8
1995	Transition/unclassified	-0.44	8	La Niña	-0.71	8
1996	Neutral	-0.29	8	Transition/unclassified	-0.10	8
1997	El Niño	1.49	8	El Niño	1.73	8
1998	La Niña	-0.44	8	La Niña	-1.14	8
1999	La Niña	-0.95	8	La Niña	-1.15	8
2000	La Niña	-0.59	8	La Niña	-0.54	8
2001	Neutral	-0.17	8	Neutral	-0.04	8
2002	El Niño	0.65	8	El Niño	0.62	8
2003	Neutral	0.07	8	Neutral	0.23	8
2004	Transition/unclassified	0.39	8	El Niño	0.53	8
2005	Transition/unclassified	-0.04	8	La Niña	-0.57	8
2006	Transition/unclassified	0.24	8	Transition/unclassified	0.33	8
2007	La Niña	-0.79	8	La Niña	-1.41	8
2008	Transition/unclassified	-0.51	8	La Niña	-0.55	8
2009	El Niño	0.50	8	El Niño	0.94	8
2010	La Niña	-0.90	8	La Niña	-1.12	8
2011	La Niña	-0.64	8	La Niña	-0.62	8
2012	Neutral	0.12	8	Neutral	-0.10	8
2013	Neutral	-0.26	8	Neutral	-0.00	8
2014	Transition/unclassified	0.31	8	Not analysed	NE	NE

Note. Rainy is May-October. Hot/dry is November-April and is labelled by its November start year. Hot/dry climate year 2014 ends outside the 1981-2014 daily analysis window and is not analysed.

Supplementary Table S4. Model-specific management-season ENSO sample sizes.

Model	Season	El Niño	Neutral	La Niña	Transition/unclassified
ACCESS-ESM1-5	Hot/dry	10	12	8	3
ACCESS-ESM1-5	Rainy	4	13	2	15
CESM2	Hot/dry	14	5	11	3
CESM2	Rainy	11	4	5	14
CanESM5	Hot/dry	6	9	5	13
CanESM5	Rainy	5	10	7	12
EC-Earth3	Hot/dry	9	11	10	3
EC-Earth3	Rainy	6	10	4	14
FGOALS-g3	Hot/dry	13	6	11	3
FGOALS-g3	Rainy	6	4	6	18
MIROC6	Hot/dry	11	5	12	5
MIROC6	Rainy	12	4	12	6
MRI-ESM2-0	Hot/dry	12	4	15	2
MRI-ESM2-0	Rainy	7	3	12	12

Note. Each GCM is classified from its own exact-member Niño-3.4 index. Persistent episodes require at least five overlapping three-month windows at |anomaly| $\geq 0.5^{\circ}\text{C}$; season assignment uses a strict majority.

Supplementary Table S5. Model-level raw and blocked cross-fitted-QDM primary rainfall responses relative to Neutral.

Season	Metric	Phase	Model	Raw response (%)	QDM response (%)	QDM-raw (pp)	Raw grids	Phase/Neutral n
Hot/dry	CDD	El Niño	ACCESS-ESM1-5	4.8	13.3	8.4	2	10/12
Hot/dry	CDD	El Niño	CESM2	-13.9	-1.9	12.0	3	14/5
Hot/dry	CDD	El Niño	CanESM5	36.5	4.9	-31.6	1	6/9
Hot/dry	CDD	El Niño	EC-Earth3	31.5	7.2	-24.3	5	9/11
Hot/dry	CDD	El Niño	FGOALS-g3	-7.2	29.6	36.8	2	13/6
Hot/dry	CDD	El Niño	MIROC6	-19.3	16.1	35.4	2	11/5
Hot/dry	CDD	El Niño	MRI-ESM2-0	18.3	13.7	-4.6	3	12/4
Hot/dry	CDD	La Niña	ACCESS-ESM1-5	-3.9	-12.1	-8.2	2	8/12
Hot/dry	CDD	La Niña	CESM2	-26.5	0.6	27.0	3	11/5
Hot/dry	CDD	La Niña	CanESM5	-6.2	-18.6	-12.4	1	5/9
Hot/dry	CDD	La Niña	EC-Earth3	-1.1	32.9	34.0	5	10/11
Hot/dry	CDD	La Niña	FGOALS-g3	-1.0	15.4	16.3	2	11/6
Hot/dry	CDD	La Niña	MIROC6	-20.3	-9.7	10.6	2	12/5
Hot/dry	CDD	La Niña	MRI-ESM2-0	7.5	6.3	-1.1	3	15/4
Hot/dry	PRCPTOT	El Niño	ACCESS-ESM1-5	-10.4	-29.4	-19.1	2	10/12
Hot/dry	PRCPTOT	El Niño	CESM2	12.3	28.9	16.6	3	14/5
Hot/dry	PRCPTOT	El Niño	CanESM5	-7.2	32.4	39.6	1	6/9
Hot/dry	PRCPTOT	El Niño	EC-Earth3	-37.3	-42.3	-5.0	5	9/11
Hot/dry	PRCPTOT	El Niño	FGOALS-g3	-24.3	32.6	56.9	2	13/6
Hot/dry	PRCPTOT	El Niño	MIROC6	-14.2	14.4	28.6	2	11/5
Hot/dry	PRCPTOT	El Niño	MRI-ESM2-0	-1.8	-22.1	-20.4	3	12/4
Hot/dry	PRCPTOT	La Niña	ACCESS-ESM1-5	-7.7	-29.5	-21.8	2	8/12
Hot/dry	PRCPTOT	La Niña	CESM2	22.3	46.0	23.6	3	11/5
Hot/dry	PRCPTOT	La Niña	CanESM5	-1.2	20.3	21.5	1	5/9
Hot/dry	PRCPTOT	La Niña	EC-Earth3	-31.5	-36.5	-5.0	5	10/11
Hot/dry	PRCPTOT	La Niña	FGOALS-g3	16.0	49.8	33.8	2	11/6
Hot/dry	PRCPTOT	La Niña	MIROC6	54.4	110.3	55.9	2	12/5
Hot/dry	PRCPTOT	La Niña	MRI-ESM2-0	-0.4	-17.4	-17.0	3	15/4
Hot/dry	Rx1day	El Niño	ACCESS-ESM1-5	-18.2	-39.2	-21.0	2	10/12
Hot/dry	Rx1day	El Niño	CESM2	11.3	-2.1	-13.3	3	14/5
Hot/dry	Rx1day	El Niño	CanESM5	13.3	62.4	49.1	1	6/9
Hot/dry	Rx1day	El Niño	EC-Earth3	-17.1	-19.1	-2.0	5	9/11
Hot/dry	Rx1day	El Niño	FGOALS-g3	-20.2	45.1	65.3	2	13/6
Hot/dry	Rx1day	El Niño	MIROC6	-4.3	51.4	55.7	2	11/5
Hot/dry	Rx1day	El Niño	MRI-ESM2-0	-22.1	-51.9	-29.8	3	12/4
Hot/dry	Rx1day	La Niña	ACCESS-ESM1-5	-31.5	-46.8	-15.3	2	8/12
Hot/dry	Rx1day	La Niña	CESM2	34.1	23.0	-11.1	3	11/5
Hot/dry	Rx1day	La Niña	CanESM5	6.2	46.2	40.0	1	5/9
Hot/dry	Rx1day	La Niña	EC-Earth3	-34.4	-21.0	13.4	5	10/11
Hot/dry	Rx1day	La Niña	FGOALS-g3	35.9	86.8	50.9	2	11/6
Hot/dry	Rx1day	La Niña	MIROC6	28.2	87.1	58.9	2	12/5
Hot/dry	Rx1day	La Niña	MRI-ESM2-0	-25.6	-54.4	-28.8	3	15/4
Hot/dry	Wet-day frequency	El Niño	ACCESS-ESM1-5	-4.4	-1.2	3.1	2	10/12
Hot/dry	Wet-day frequency	El Niño	CESM2	3.8	9.2	5.4	3	14/5
Hot/dry	Wet-day frequency	El Niño	CanESM5	-4.8	-11.8	-7.1	1	6/9
Hot/dry	Wet-day frequency	El Niño	EC-Earth3	-16.1	-28.5	-12.5	5	9/11
Hot/dry	Wet-day frequency	El Niño	FGOALS-g3	-14.4	1.5	15.8	2	13/6
Hot/dry	Wet-day frequency	El Niño	MIROC6	-2.9	-1.7	1.2	2	11/5
Hot/dry	Wet-day frequency	El Niño	MRI-ESM2-0	-6.8	-6.2	0.6	3	12/4
Hot/dry	Wet-day frequency	La Niña	ACCESS-ESM1-5	4.7	11.5	6.7	2	8/12
Hot/dry	Wet-day frequency	La Niña	CESM2	7.8	35.3	27.5	3	11/5
Hot/dry	Wet-day frequency	La Niña	CanESM5	5.4	20.7	15.3	1	5/9
Hot/dry	Wet-day frequency	La Niña	EC-Earth3	-11.8	-19.5	-7.7	5	10/11
Hot/dry	Wet-day frequency	La Niña	FGOALS-g3	0.1	5.7	5.7	2	11/6
Hot/dry	Wet-day frequency	La Niña	MIROC6	19.3	49.5	30.2	2	12/5
Hot/dry	Wet-day frequency	La Niña	MRI-ESM2-0	-7.7	3.5	11.2	3	15/4
Rainy	CDD	El Niño	ACCESS-ESM1-5	20.7	36.2	15.6	2	4/13
Rainy	CDD	El Niño	CESM2	40.3	11.9	-28.3	3	11/4
Rainy	CDD	El Niño	CanESM5	44.0	52.5	8.5	1	5/10
Rainy	CDD	El Niño	EC-Earth3	10.1	13.6	3.5	5	6/10
Rainy	CDD	El Niño	FGOALS-g3	4.9	-3.7	-8.6	2	6/4

Season	Metric	Phase	Model	Raw response (%)	QDM response (%)	QDM-raw (pp)	Raw grids	Phase/Neutral n
Rainy	CDD	El Niño	MIROC6	-4.3	7.0	11.3	2	12/4
Rainy	CDD	El Niño	MRI-ESM2-0	13.1	-19.0	-32.1	3	7/3
Rainy	CDD	La Niña	CESM2	17.1	-7.8	-25.0	3	5/4
Rainy	CDD	La Niña	CanESM5	69.5	63.4	-6.1	1	7/10
Rainy	CDD	La Niña	EC-Earth3	18.2	18.6	0.4	5	4/10
Rainy	CDD	La Niña	FGOALS-g3	15.3	-5.4	-20.7	2	6/4
Rainy	CDD	La Niña	MIROC6	15.2	12.2	-3.0	2	12/4
Rainy	CDD	La Niña	MRI-ESM2-0	6.9	-8.3	-15.2	3	12/3
Rainy	PRCPTOT	El Niño	ACCESS-ESM1-5	23.8	26.4	2.6	2	4/13
Rainy	PRCPTOT	El Niño	CESM2	-14.7	-9.6	5.1	3	11/4
Rainy	PRCPTOT	El Niño	CanESM5	-20.1	-30.6	-10.4	1	5/10
Rainy	PRCPTOT	El Niño	EC-Earth3	8.2	1.5	-6.7	5	6/10
Rainy	PRCPTOT	El Niño	FGOALS-g3	-6.8	-14.5	-7.7	2	6/4
Rainy	PRCPTOT	El Niño	MIROC6	-24.6	-18.0	6.6	2	12/4
Rainy	PRCPTOT	El Niño	MRI-ESM2-0	-12.2	-24.6	-12.3	3	7/3
Rainy	PRCPTOT	La Niña	CESM2	-9.5	-4.1	5.3	3	5/4
Rainy	PRCPTOT	La Niña	CanESM5	-20.7	-32.6	-11.9	1	7/10
Rainy	PRCPTOT	La Niña	EC-Earth3	-0.2	-6.4	-6.1	5	4/10
Rainy	PRCPTOT	La Niña	FGOALS-g3	-19.0	-27.8	-8.8	2	6/4
Rainy	PRCPTOT	La Niña	MIROC6	3.7	11.0	7.3	2	12/4
Rainy	PRCPTOT	La Niña	MRI-ESM2-0	-10.6	-23.7	-13.1	3	12/3
Rainy	Rx1day	El Niño	ACCESS-ESM1-5	-9.1	9.9	19.1	2	4/13
Rainy	Rx1day	El Niño	CESM2	-28.6	-3.8	24.8	3	11/4
Rainy	Rx1day	El Niño	CanESM5	-28.3	-39.6	-11.3	1	5/10
Rainy	Rx1day	El Niño	EC-Earth3	15.7	-5.7	-21.3	5	6/10
Rainy	Rx1day	El Niño	FGOALS-g3	-1.6	4.9	6.5	2	6/4
Rainy	Rx1day	El Niño	MIROC6	-25.3	-15.5	9.8	2	12/4
Rainy	Rx1day	El Niño	MRI-ESM2-0	-12.7	-47.9	-35.1	3	7/3
Rainy	Rx1day	La Niña	CESM2	-24.2	-24.5	-0.3	3	5/4
Rainy	Rx1day	La Niña	CanESM5	-56.9	-53.8	3.1	1	7/10
Rainy	Rx1day	La Niña	EC-Earth3	-4.7	-10.4	-5.7	5	4/10
Rainy	Rx1day	La Niña	FGOALS-g3	-20.9	-45.7	-24.7	2	6/4
Rainy	Rx1day	La Niña	MIROC6	-4.9	21.2	26.1	2	12/4
Rainy	Rx1day	La Niña	MRI-ESM2-0	5.2	-32.1	-37.4	3	12/3
Rainy	Wet-day frequency	El Niño	ACCESS-ESM1-5	1.8	-5.7	-7.5	2	4/13
Rainy	Wet-day frequency	El Niño	CESM2	-4.4	-9.2	-4.8	3	11/4
Rainy	Wet-day frequency	El Niño	CanESM5	-7.3	-15.3	-8.0	1	5/10
Rainy	Wet-day frequency	El Niño	EC-Earth3	5.5	-3.0	-8.5	5	6/10
Rainy	Wet-day frequency	El Niño	FGOALS-g3	2.3	-4.0	-6.3	2	6/4
Rainy	Wet-day frequency	El Niño	MIROC6	-8.2	-10.0	-1.9	2	12/4
Rainy	Wet-day frequency	El Niño	MRI-ESM2-0	-5.6	-7.0	-1.4	3	7/3
Rainy	Wet-day frequency	La Niña	CESM2	3.5	3.2	-0.2	3	5/4
Rainy	Wet-day frequency	La Niña	CanESM5	-7.9	-15.9	-7.9	1	7/10
Rainy	Wet-day frequency	La Niña	EC-Earth3	-12.1	-7.1	5.0	5	4/10
Rainy	Wet-day frequency	La Niña	FGOALS-g3	-19.6	-4.4	15.2	2	6/4
Rainy	Wet-day frequency	La Niña	MIROC6	-3.2	-2.8	0.4	2	12/4
Rainy	Wet-day frequency	La Niña	MRI-ESM2-0	-5.6	-5.0	0.6	3	12/3

Note. Responses are model medians after collapsing stations sharing an identical raw-grid signature. A minimum of three phase and three Neutral seasons is required; unavailable model-phase combinations are omitted. pp = percentage points.

Supplementary Table S6. Descriptive secondary-index responses relative to Neutral for observations, raw CMIP6, and cross-fitted QDM.

Season	Metric	Phase	Observed response (%)	Observed stations	Raw ensemble (%)	QDM ensemble (%)	Raw/QDM models
Hot/dry	CWD	El Niño	-22.7	13	-14.7	-0.3	7/7
Hot/dry	CWD	La Niña	28.8	13	-5.9	17.3	7/7
Hot/dry	R10mm	El Niño	-36.1	13	-26.6	7.6	7/7
Hot/dry	R10mm	La Niña	11.4	13	21.2	40.5	7/7
Hot/dry	R20mm	El Niño	-44.0	13	-12.5	34.4	7/7
Hot/dry	R20mm	La Niña	8.5	13	47.3	17.0	7/7
Hot/dry	R50mm	El Niño	-42.9	13	-77.3	-2.0	7/7
Hot/dry	R50mm	La Niña	-20.0	13	-52.3	-9.7	7/7
Hot/dry	R95p	El Niño	-43.3	13	-30.0	40.0	7/7
Hot/dry	R95p	La Niña	-18.3	13	8.0	38.7	7/7
Hot/dry	R99p	El Niño	-70.2	13	-14.1	-47.5	7/7
Hot/dry	R99p	La Niña	-37.6	13	-50.7	-34.2	7/7
Hot/dry	Rx5day	El Niño	-39.2	13	-19.4	7.6	7/7
Hot/dry	Rx5day	La Niña	1.9	13	1.1	16.6	7/7
Hot/dry	q50	El Niño	-15.7	12	-2.0	11.8	7/7
Hot/dry	q50	La Niña	-3.1	13	3.8	13.3	7/7
Hot/dry	q90	El Niño	-27.3	12	-11.2	4.9	7/7
Hot/dry	q90	La Niña	-1.2	13	6.2	16.0	7/7
Hot/dry	q95	El Niño	-24.0	12	-6.7	23.7	7/7
Hot/dry	q95	La Niña	-13.7	13	14.0	16.1	7/7
Hot/dry	q99	El Niño	-20.7	12	-11.3	7.9	7/7
Hot/dry	q99	La Niña	-11.4	13	2.3	24.6	7/7
Rainy	CWD	El Niño	-9.1	13	-5.1	5.1	7/7
Rainy	CWD	La Niña	10.2	13	-1.9	-1.2	6/6
Rainy	R10mm	El Niño	-5.0	13	-8.1	-10.7	7/7
Rainy	R10mm	La Niña	10.4	13	-4.9	-11.5	6/6
Rainy	R20mm	El Niño	-16.7	13	-17.5	-14.9	7/7
Rainy	R20mm	La Niña	7.6	13	-27.0	-9.5	6/6
Rainy	R50mm	El Niño	-40.0	13	-52.6	-41.2	7/7
Rainy	R50mm	La Niña	2.9	13	-67.3	-35.6	6/6
Rainy	R95p	El Niño	-28.5	13	-36.7	-22.2	7/7
Rainy	R95p	La Niña	-0.0	13	-33.9	-36.6	6/6
Rainy	R99p	El Niño	-48.3	13	-45.2	-19.9	7/7
Rainy	R99p	La Niña	47.1	13	-32.9	-52.6	6/6
Rainy	Rx5day	El Niño	-10.4	13	-21.9	-3.1	7/7
Rainy	Rx5day	La Niña	13.7	13	-15.8	-30.7	6/6
Rainy	q50	El Niño	-11.7	13	0.8	-7.8	7/7
Rainy	q50	La Niña	10.8	13	6.6	-4.9	6/6
Rainy	q90	El Niño	-6.8	13	-1.1	-9.0	7/7
Rainy	q90	La Niña	8.5	13	0.9	-11.9	6/6
Rainy	q95	El Niño	-7.9	13	-5.6	-9.8	7/7
Rainy	q95	La Niña	4.0	13	-7.5	-10.7	6/6
Rainy	q99	El Niño	-19.3	13	-13.0	-8.0	7/7
Rainy	q99	La Niña	5.4	13	-14.9	-25.9	6/6

Note. Secondary indices were not included in the primary Benjamini-Hochberg family and are presented without inferential claims. NE = not estimable because the Neutral reference was zero or too few eligible entities were available. Definitions: Rx5day, maximum five-day total; CWD, maximum wet spell; R10/20/50mm, counts above thresholds; R95p/R99p, totals above observed wet-day percentiles; q50/q90/q95/q99, wet-day quantiles.

Supplementary Table S7. Phase contrast and true neutral-centred asymmetry with event-level inference.

Source	Season	Metric	Phase contrast % [95% CI]	p	True asymmetry % [95% CI]	p	BH q
Observed	Hot/dry	CDD	-3.6 [-25.8, 13.5]	0.800	-24.4 [-62.1, 18.2]	0.343	0.781
Observed	Hot/dry	PRCPTOT	46.0 [15.1, 82.7]	0.014	-15.1 [-61.9, 52.5]	0.677	0.781
Observed	Hot/dry	Rx1day	20.5 [-1.8, 48.1]	0.217	-23.9 [-67.7, 27.2]	0.438	0.781
Observed	Hot/dry	Wet-day frequency	43.8 [9.9, 69.1]	0.003	15.5 [-31.3, 58.7]	0.562	0.781
Observed	Rainy	CDD	-19.4 [-40.7, -1.4]	0.064	8.8 [-20.1, 48.5]	0.625	0.781
Observed	Rainy	PRCPTOT	11.1 [-4.2, 23.4]	0.125	-3.5 [-25.4, 16.1]	0.752	0.820
Observed	Rainy	Rx1day	14.8 [-2.9, 51.0]	0.420	-23.0 [-58.4, 37.4]	0.451	0.781
Observed	Rainy	Wet-day frequency	2.0 [-5.8, 10.4]	0.499	-6.1 [-20.6, 7.3]	0.304	0.781
Qdm	Hot/dry	CDD	-14.2 [-30.1, 4.7]	0.046	6.4 [-12.5, 42.2]	0.683	0.781
Qdm	Hot/dry	PRCPTOT	5.5 [-10.6, 38.4]	0.679	52.7 [-37.8, 117.0]	0.068	0.409
Qdm	Hot/dry	Rx1day	-1.5 [-14.9, 34.8]	0.917	20.9 [-47.0, 144.0]	0.501	0.781
Qdm	Hot/dry	Wet-day frequency	12.7 [1.1, 32.2]	0.064	8.8 [-13.6, 40.4]	0.521	0.781
Qdm	Rainy	CDD	1.7 [-13.0, 18.5]	0.798	7.6 [-7.9, 38.0]	0.571	0.781
Qdm	Rainy	PRCPTOT	-0.6 [-10.3, 11.2]	0.924	-28.0 [-45.6, -5.2]	0.068	0.409
Qdm	Rainy	Rx1day	-7.0 [-22.7, 9.8]	0.615	-34.6 [-67.4, 11.9]	0.290	0.781
Qdm	Rainy	Wet-day frequency	-0.4 [-3.1, 8.8]	0.873	-10.6 [-18.0, -3.9]	0.058	0.409
Raw	Hot/dry	CDD	-12.6 [-26.4, 1.7]	0.066	1.0 [-22.6, 29.1]	0.944	0.959
Raw	Hot/dry	PRCPTOT	6.0 [-2.9, 27.5]	0.446	-8.3 [-34.8, 36.7]	0.586	0.781
Raw	Hot/dry	Rx1day	2.3 [-15.6, 26.2]	0.832	15.7 [-40.7, 48.3]	0.450	0.781
Raw	Hot/dry	Wet-day frequency	9.1 [0.3, 16.0]	0.018	0.4 [-18.4, 13.2]	0.959	0.959
Raw	Rainy	CDD	7.0 [-12.4, 26.8]	0.415	22.8 [9.3, 80.1]	0.203	0.697
Raw	Rainy	PRCPTOT	0.6 [-7.9, 14.4]	0.935	-23.4 [-37.7, -5.7]	0.123	0.590
Raw	Rainy	Rx1day	-7.5 [-17.5, 14.8]	0.436	-26.4 [-55.4, 1.5]	0.174	0.697
Raw	Rainy	Wet-day frequency	-0.5 [-7.0, 6.4]	0.882	-12.2 [-21.1, 1.5]	0.058	0.409

Note. Phase contrast = ALa - AEI. True neutral-centred asymmetry = ALa + AEI. Bootstrap repetitions = 5,000; permutation repetitions = 4,999; resampling unit = management-season climate year.

Supplementary Table S8. Cross-fitted-QDM ENSO signal-preservation diagnostics.

Season	Metric	Phase	Observed (%)	Raw (%)	QDM (%)	Category	Ratio	Shift (pp)	Direction agreement raw/QDM	Paired sign retention
Hot/dry	CDD	El Niño	-9.6	4.8	13.3	Indeterminate	NE	8.4	0.43/0.14	0.71 (7 models)
Hot/dry	CDD	La Niña	-18.5	-3.9	0.6	Indeterminate	NE	4.4	0.86/0.43	0.57 (7 models)
Hot/dry	PRCPTOT	El Niño	-28.7	-10.4	14.4	REVERSED	1.39	24.8	0.86/0.43	0.57 (7 models)
Hot/dry	PRCPTOT	La Niña	13.3	-0.4	20.3	Indeterminate	NE	20.7	0.43/0.57	0.86 (7 models)
Hot/dry	Rx1day	El Niño	-27.2	-17.1	-2.1	ATTENUATED	0.12	15.0	0.71/0.57	0.57 (7 models)
Hot/dry	Rx1day	La Niña	-12.6	6.2	23.0	AMPLIFIED	3.71	16.8	0.43/0.43	1.00 (7 models)
Hot/dry	Wet-day frequency	El Niño	-10.4	-4.8	-1.7	Indeterminate	NE	3.1	0.86/0.71	0.86 (7 models)
Hot/dry	Wet-day frequency	La Niña	33.0	4.7	11.5	Indeterminate	NE	6.7	0.71/0.86	0.86 (7 models)
Rainy	CDD	El Niño	17.9	13.1	11.9	PRESERVED	0.91	-1.1	0.86/0.71	0.57 (7 models)
Rainy	CDD	La Niña	-5.8	16.2	3.4	ATTENUATED	0.21	-12.8	0.00/0.50	0.50 (6 models)
Rainy	PRCPTOT	El Niño	-7.2	-12.2	-14.5	PRESERVED	1.19	-2.3	0.71/0.71	1.00 (7 models)
Rainy	PRCPTOT	La Niña	2.7	-10.0	-15.0	AMPLIFIED	1.50	-5.0	0.17/0.17	1.00 (6 models)
Rainy	Rx1day	El Niño	-24.3	-12.7	-5.7	ATTENUATED	0.44	7.1	0.86/0.71	0.57 (7 models)
Rainy	Rx1day	La Niña	0.5	-12.9	-28.3	AMPLIFIED	2.19	-15.4	0.17/0.17	0.67 (6 models)
Rainy	Wet-day frequency	El Niño	-0.1	-4.4	-7.0	Indeterminate	NE	-2.6	0.57/1.00	0.57 (7 models)
Rainy	Wet-day frequency	La Niña	-2.4	-6.8	-4.7	ATTENUATED	0.69	2.1	0.83/0.83	1.00 (6 models)

Note. A raw response with magnitude <5 percentage points is indeterminate. Otherwise: reversed = sign changed; attenuated = $|QDM/raw| < 0.8$; preserved = 0.8-1.2; amplified = > 1.2 .

Supplementary Table S9. Blocked cross-fitted-QDM calibration and leakage diagnostic summary by model.

Model	Member	Fits	Blocks	Fallback fits	Target observations used	Min observed wet	Min model wet	Target n range
ACCESS-ESM1-5	r1i1p1f1	1092	7	82	0	30	30	84-155
CESM2	r1i1p1f1	1092	7	82	0	30	30	84-155
CanESM5	r1i1p1f1	1092	7	82	0	30	30	84-155
EC-Earth3	r1i1p1f1	1092	7	82	0	30	30	84-155
FGOALS-g3	r1i1p1f1	1092	7	82	0	30	30	84-155
MIROC6	r1i1p1f1	1092	7	82	0	30	30	84-155
MRI-ESM2-0	r1i1p1f1	1092	7	82	0	30	30	84-155

Note. All 7,644 fits used zero observations from their target fold. The adjacent-three-month calibration pool was activated only when a single calendar month failed the ≥ 30 wet-day calibration requirement; the target remained in its original month.

Supplementary Table S10a. Pre-specified analysis acceptance gates.

Gate	Description	Status
P3-A	Observed and model ENSO sources are frozen and traceable	PASS
P3-B	Only complete common-365 station-seasons enter metrics	PASS
P3-C	All season ENSO labels use the frozen allowed vocabulary	PASS
P3-D	Common period contains 13 gauges and seven models	PASS
P3-E	Blocked cross-fitted QDM used no target observations	PASS
P3-F	Both phase-vs-neutral primary responses exist	PASS
P3-G	Phase contrast and true neutral-centred asymmetry are separated	PASS
P3-H	Pre-specified event-level bootstrap and permutation inference completed	PASS

Supplementary Table S10b. Machine-readable supplementary artifacts and their analytical roles.

File	Rows	Role
enso_episode_catalog_observed.csv	45	Observed persistent ONI episodes and intensity summaries
enso_episode_catalog_models.csv	128	Exact-member persistent Niño-3.4 episodes
season_classification_observed.csv	68	Observed rainy and hot/dry season labels
season_classification_models.csv	476	Model-specific management-season labels
observed_station_primary_responses.csv	208	Gauge-level primary phase responses
model_primary_responses.csv	216	Raw and QDM model-level primary responses
primary_response_summary.csv	48	Observed/raw/QDM ensemble inference
enso_asymmetry_primary.csv	24	Phase contrast and true asymmetry inference
qdm_enso_signal_preservation.csv	16	Preservation classification and directional agreement
crossfit_qdm_diagnostics.csv	7644	Fold/month calibration and leakage audit
Paper3_Supplementary_Secondary_Responses.csv	44	Descriptive secondary-index responses generated for Table S6
PAPER3_ACCEPTANCE_GATES.csv	8	Pre-specified reproducibility gates

Note. Paths, checksums, source capture metadata, the executable notebook, and the complete daily/seasonal intermediate outputs are retained in the reproducibility package. Station observations remain subject to provider terms.